

Takyoung Kim

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Interests

My research interest lies in Natural Language Processing, Spoken Language Understanding, and Conversational AI. I am particularly interested in developing companion AI systems that can (1) understand social and cognitive context beyond the textual domain, (2) provide personalized and adaptive assistance, and (3) seamlessly interact within the physical world.

Education

University of Illinois Urbana-Champaign, Urbana, IL, USA 🔗 Doctor of Philosophy, Computer Science Advisor: Dilek Hakkani-Tür	Aug 2024 – Present
Korea University, Seoul, South Korea 🔗 Master of Engineering, Industrial Management Engineering Advisor: Pilsung Kang (currently affiliated with Seoul National University)	Sep 2020 – Aug 2022
Sungkyunkwan University, Seoul, South Korea 🔗 Bachelor of Science in Engineering, Interdisciplinary Program of Convergent Software Bachelor of Media & Communication (<i>Leave of absence for military service: Feb 2015 - Nov 2016</i>)	Mar 2013 – Aug 2020

Employment

Meta Superintelligence Labs Voice Modeling Team Menlo Park, CA, USA 🔗 <i>Research Scientist Intern</i> (Mentors: Naoyuki Kanda , Abdelrahman Mohamed) Working on enhancing spoken education systems	May 2026 – Present
Amazon Alexa AI Conversational Assistant Modeling and Learning Bellevue, WA, USA 🔗 <i>Applied Scientist Intern</i> (Mentors: Jinseok Nam , Chandrayee Basu , Chengyuan Ma) Worked on error recovery in conversational systems with deployment constraints [3]	May 2025 – Aug 2025
Korea University Data Science & Business Analytics Lab Seoul, South Korea 🔗 <i>Research Associate</i> (Advisor: Pilsung Kang) Worked on retrieval-augmented language models specialized in Korean digital media	Apr 2024 – Jul 2024
LG AI Research Advanced Machine Learning Lab Seoul, South Korea 🔗 <i>Research Scientist Intern</i> (Mentors: Kyungjae Lee , Moontae Lee) Worked on improving long-form retrieval-augmented generation via planning [7]	Apr 2023 – Mar 2024
NAVER AI Lab Language Research Team Seongnam, South Korea 🔗 <i>Research Scientist Intern</i> (Mentor: Sungdong Kim) Worked on revealing innate bias in task-oriented dialogue research via interactive user study [8] Worked on constructing safety benchmark for sensitive questions [11] and social biases [10]	Jul 2022 – Jan 2023

Honors and Awards

Conference Travel Grant (Registration + 1,600 USD) ICLR 2026 [3]	Apr 2026
CS PhD Fellowship University of Illinois Urbana-Champaign	Aug 2024 – May 2025
Best Paper Award ACL 2024 Workshop on NLP for Conversational AI [8]	Aug 2024
Best Student Feedback Award, Research Ideation Competition Korea University	Feb 2021, Dec 2021
2nd Place (2,500 USD), MyData Acting Hackathon Ministry of Science and ICT	Oct 2019
Merit-based Scholarship, Computer Science Coursework Sungkyunkwan University	Spring 2018, Fall 2018

Published Papers

- [1] **AURA: A Diagnostic Framework for Tracking User Satisfaction of Interactive Planning Agents** 
Takyoun Kim^{*}, Janvijay Singh^{*}, Shuhaib Mehri^{*}, Emre Can Acikgoz, Sagnik Mukherjee, Nimet Beyza Bozdog, Sumuk Shashidhar, Gokhan Tur, Dilek Hakkani-Tür
IEEE Transactions on Audio, Speech and Language Processing (Q1, Impact Factor: 5.1) IEEE TASLP 2026
Workshop on Multi-Turn Interactions in Large Language Models MTI-LLM, NeurIPS 2025
- [2] **From Documents to Segments: A Contextual Reformulation for Topic Assignment** 
 Hoonsang Yoon^{*}, Takyoun Kim^{*}, Wonkee Lee, Ilmin Cho, Dilek Hakkani-Tür[†], Stanley Jungkyu Choi[†]
Annual Meeting of the Association for Computational Linguistics Findings of ACL 2026
- [3] **ReIn: Conversational Error Recovery with Reasoning Inception** 
Takyoun Kim, Jinseok Nam, Chandrayee Basu, Xing Fan, Chengyuan Ma, Heng Ji, Gokhan Tur, Dilek Hakkani-Tür
International Conference on Learning Representations ICLR 2026
- [4] **Goal Alignment in LLM-Based User Simulators for Conversational AI** 
 Shuhaib Mehri, Xiaocheng Yang, Takyoun Kim, Gokhan Tur, Shikib Mehri, Dilek Hakkani-Tür
Transactions of the Association for Computational Linguistics (Oral Presentation at ACL 2026) TACL 2026
- [5] **TD-EVAL: Revisiting Task-Oriented Dialogue Evaluation by Combining Turn-Level Precision with Dialogue-Level Comparisons** 
 Emre Can Acikgoz^{*}, Carl Guo^{*}, Suvodip Dey^{*}, Akul Datta, Takyoun Kim, Gokhan Tur, Dilek Hakkani-Tür
Annual Meeting of the Special Interest Group on Discourse and Dialogue (Oral Presentation) SIGDIAL 2025
- [6] **Premise-Augmented Reasoning Chains Improve Error Identification in Math Reasoning with LLMs** 
 Sagnik Mukherjee^{*}, Abhinav Chinta^{*}, Takyoun Kim, Tarun Anoop Sharma, Dilek Hakkani-Tür
International Conference on Machine Learning ICML 2025
- [7] **Learning to Explore and Select for Coverage-Conditioned Retrieval-Augmented Generation** 
Takyoun Kim, Kyungjae Lee, Young Rok Jang, Ji Yong Cho, Gangwoo Kim, Minseok Cho, Moontae Lee
Nations of the Americas Chapter of the Association for Computational Linguistics Findings of NAACL 2025
- [8] **Revealing User Familiarity Bias in Task-Oriented Dialogue via Interactive Evaluation** 
Takyoun Kim, Jamin Shin, Young-Ho Kim, Sanghwan Bae, Sungdong Kim
Workshop on NLP for Conversational AI (Oral Presentation, Best Paper Award) NLP4ConvAI, ACL 2024
- [9] **DSTEa: Improving Dialogue State Tracking via Entity Adaptive Pre-training** 
 Yukyung Lee, Takyoun Kim, Hoonsang Yoon, Pilsung Kang, Junseong Bang, Misuk Kim
Knowledge-Based Systems (Q1, Impact Factor: 8.8) KBS 2024
Workshop on Knowledge Augmented Methods for NLP KnowledgeNLP, KDD 2023
- [10] **KoSBi: A Dataset for Mitigating Social Bias Risks Towards Safer Large Language Model Applications** 
 Hwaran Lee^{*}, Seokhee Hong^{*}, Joonsuk Park, Takyoun Kim, Gunhee Kim, Jung-Woo Ha
Annual Meeting of the Association for Computational Linguistics (Industry Track) ACL 2023
- [11] **SQuARe: A Large-Scale Dataset of Sensitive Questions and Acceptable Responses Created Through Human-Machine Collaboration** 
 Hwaran Lee^{*}, Seokhee Hong^{*}, Joonsuk Park, Takyoun Kim, Meeyoung Cha, Yejin Choi, Byoungpil Kim, Gunhee Kim, Eun-Ju Lee, Yong Lim, Alice Oh, Sangchul Park, Jung-Woo Ha
Annual Meeting of the Association for Computational Linguistics (Oral Presentation, Best Paper Nomination) ACL 2023
- [12] **Oh My Mistake!: Toward Realistic Dialogue State Tracking including Turnback Utterances** 
Takyoun Kim, Yukyung Lee, Hoonsang Yoon, Pilsung Kang, Junseong Bang, Misuk Kim
Towards Semi-Supervised and Reinforced Task-Oriented Dialog Systems SereTOD, EMNLP 2022
- [13] **Mismatch between Multi-turn Dialogue and its Evaluation Metric in Dialogue State Tracking** 
Takyoun Kim, Hoonsang Yoon, Yukyung Lee, Pilsung Kang, Misuk Kim
Annual Meeting of the Association for Computational Linguistics (Short) ACL 2022
- [14] **Building an Integrated Framework of Korean Text Summarization and Text-to-Speech** 
Takyoun Kim, Jina Kim, Hyeongwon Kang, Subin Kim, Pilsung Kang
Journal of Korean Institute of Industrial Engineers KIIIE 2022

Presentations and Talks

DuplexGen: Adaptive Synthesis of Human-AI Turn-Taking Dialogues Midwest Speech and Language Days (Urbana, IL, USA)	Apr 2026
Seamless Spoken Interaction with AI Companions: Naturalness and Efficiency PhD Qualifying Exam (Subarea: AI, Committees: <i>Minje Kim, Nancy Amato, Tong Zhang</i>)	Feb 2026
Data-Driven Approaches to Personalizing Spoken Interaction in Conversational AI Seoul National University (Seoul, South Korea), Host: <i>Pilsung Kang</i> Hanyang University (Seoul, South Korea), Host: <i>Misuk Kim</i>	Dec 2025 Dec 2025
Explainable and Efficient Turn-Taking Strategies for Full-Duplex Conversational AI TTIC Summer Workshop on Foundations of Speech and Audio Foundation Models (Chicago, IL, USA)	Sep 2025
Embodied Multi-Agent Planning in Conflicting Scenarios Midwest Speech and Language Days (Notre Dame, IN, USA)	Apr 2025
Relaxing Unrealistic Assumptions of Dataset and Evaluation Metric in Dialogue State Tracking Master's Thesis (Committees: <i>Pilsung Kang, Seoung Bum Kim, Sung Won Han</i>)	Jun 2022

Teaching Experience

CS598 Conversational AI University of Illinois Urbana-Champaign Teaching Assistant (Instructor: <i>Dilek Hakkani-Tür</i>)	Spring 2026
T02459101 Natural Language Processing Hankuk University of Foreign Studies Teaching Assistant (Instructor: <i>Yukyung Lee</i>)	Spring 2024
Industrial Machine Learning LG Electronics Head Teaching Assistant (Instructor: <i>Pilsung Kang</i>)	Spring 2022
Research Mentoring Hyundai Mobis Research Project Assistant (Instructor: <i>Pilsung Kang</i>)	Summer 2021
Industrial Machine Learning and Deep Learning LG Innotek, SK Hynix Teaching Assistant (Instructor: <i>Pilsung Kang</i>)	Summer 2021

Academic Service

Conference Reviewer	
Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)	2026
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)	2026
International Conference on Learning Representations (ICLR)	2026
Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)	2025
Conference on Language Modeling (COLM)	2025 – 2026
Annual Meeting of the Association for Computational Linguistics (ACL)	2025 – 2026
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)	2025
European Chapter of the Association for Computational Linguistics (EACL)	2024 – 2026
Empirical Methods in Natural Language Processing (EMNLP)	2023 – 2025
Neural Information Processing Systems (NeurIPS)	2022 – 2026
Journal Reviewer	
Knowledge-Based Systems	2023 – 2024
Workshop Reviewer	
Workshop on Multi-Turn Interactions in Large Language Models @ NeurIPS	2025
Organizer	
Midwest Speech and Language Days	2026
Student Volunteer (On-Site)	
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)	2025